Exp no - 8
Date - 20/08/2026
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# NMAP TO DISCOVER LIVE HOSTS USING NMAP SCANS (ARP, ICMP, TCP/UDP) ON THE TRYHACKME PLATFORM

AIM:

To discover live hosts in a network using Nmap scans based on ARP, ICMP, TCP and UDP techniques on the TryHackMe platform and understand how Nmap identifies active systems.

ALGORITHM:

1. Open the TryHackMe website and log in to the account.
2. Open the Nmap Live Host Discovery room specified in the laboratory manual.
3. Start the provided virtual machine or target environment.
4. Open the terminal in the designated environment.
5. Identify the target IP address provided by the platform.
6. Perform an ARP scan to discover active hosts on the local network.
7. Perform an ICMP scan to check whether the target hosts respond to ICMP requests.
8. Perform TCP-based host discovery to identify active hosts using TCP probes.
9. Perform UDP-based host discovery to identify active hosts using UDP probes.
10. Observe the scan results and note the IP addresses of the discovered hosts.
11. Compare the results obtained using the different host discovery techniques.

PROGRAMS / COMMANDS:

The following Nmap commands are used to perform host discovery in the authorized TryHackMe lab environment.

1. ARP SCAN

Command:

nmap -sn -PR 192.168.1.0/24

Purpose: Discovers active hosts on a local network using ARP requests.

2. ICMP SCAN

Command:

nmap -sn -PE 192.168.1.0/24

Purpose: Uses ICMP echo requests to identify hosts that respond to network probes.

3. TCP SCAN

Command:

nmap -sn -PS80,443 192.168.1.0/24

Purpose: Uses TCP SYN probes to check whether hosts are active.

4. UDP SCAN

Command:

nmap -sn -PU40125 192.168.1.0/24

Purpose: Uses UDP probes for host discovery.

Note: Replace 192.168.1.0/24 with the target network or IP address provided by TryHackMe. Run these scans only against authorized targets.

OUTPUT:

Sample output:

Starting Nmap

Nmap scan report for 192.168.1.1
Host is up.

Nmap scan report for 192.168.1.10
Host is up.

Nmap scan report for 192.168.1.15
Host is up.

Nmap done: 256 IP addresses scanned

The scan identifies hosts that respond to the selected discovery probes. The actual results depend on the target network and its firewall configuration.

RESULT:

Thus, Nmap was used to study live host discovery using ARP, ICMP, TCP and UDP scanning techniques. The experiment demonstrated how different network probes help identify active hosts in an authorized network environment.